Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

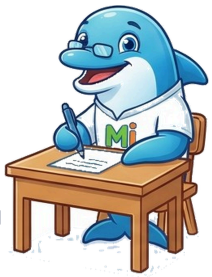
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Factor Hunters**	6-7	8-9	10 and 12	11	☐
2	**Big number puzzles**	Not in this proof edition				☐
3	**Operation detectives**	Not in this proof edition				☐
4	**Fraction frontier**	Not in this proof edition				☐
5	**Decimal detectives**	Not in this proof edition				☐
6	**Angle turns**	Not in this proof edition				☐
7	**Area architects II**	Not in this proof edition				☐
8	**Symmetry and nets**	Not in this proof edition				☐
9	**Data detectives**	Not in this proof edition				☐
10	**Puzzle day**	Not in this proof edition				☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Factors and multiples

My goal. I can list every factor of a number in order. I can find common factors and common multiples and use them to solve sharing and timetable puzzles.

A factor goes into a number exactly. A multiple is in a times table. Write factors in pairs, in order, and stop when the pairs meet.

Worked example 1

Find all the factors of 54.

Try	Pair
1	1 × 54
2	2 × 27
3	3 × 18
4, 5	no
6	6 × 9
7, 8	no

1. Try 1, 2, 3 in order. Write a pair each time the number goes in.
2. 4 and 5 do not go into 54. 6 × 9 = 54 is a pair.
3. 7 and 8 do not go in, and the next number, 9, is already used. The pairs have met, so stop.

Answer: The factors of 54 are 1, 2, 3, 6, 9, 18, 27, 54.

Worked example 2

Lamp A flashes every 4 seconds. Lamp B flashes every 10 seconds. They flash together now. After how many seconds do they next flash together?

Lamp	Flashes after these seconds
A	4, 8, 12, 16, 20
B	10, 20

1. List the multiples of each number.
2. Find the first number in both lists.

Answer: They next flash together after 20 seconds.

How to hunt

1. Factors: try 1, 2, 3 in order. Stop when the pairs meet.
2. Common factors: list both numbers, then pick the numbers in both lists.
3. Common multiples: list the multiples of both. The first number in both lists is the least.

Good to know

- A pair like 6 × 6 counts as one factor.
- So 36 has 9 factors, an odd number.

Your turn

Answer at the back of the book.

Try **Write all the factors of 45, in order.**

_____

Start here

Do these first. Take your time.

1 **Fill in the factor pairs of 56. Some numbers do not go in.**

(a) 1 × 56 2 × ☐ 4 × ☐ 7 × ☐

(b) How many factors does 56 have? ☐ factors

2 **Write all the factors of 66, in order from the smallest.**

_____

3 **A class has 32 children. The teacher makes equal teams. Every child is in a team. There is more than 1 team and more than 1 child in a team. List every team size that is possible.**

_____

Working: _____

4 **Circle True or False.**

(a) 7 is a factor of 91. True False

(b) 54 is a multiple of 8. True False

(c) 9 is a factor of 108. True False

(d) 65 is a multiple of 13. True False

5 **Write the first 6 multiples of each number. Then find the least number that is in both lists.**

(a) Multiples of 9: _____

(b) Multiples of 12: _____

(c) Least common multiple: ☐

Start here

Keep going. Check your work.

6 **Find the common factors of 28 and 42.**

(a) Factors of 28: _____

(b) Factors of 42: _____

(c) Common factors: _____

(d) Greatest common factor: ☐

7 **Ravi has 30 red marbles and 45 blue marbles. He fills equal bags and no marble is left out. Each bag holds the same number of red marbles, and the same number of blue marbles.**

(a) What is the most bags he can fill? ☐ bags

(b) Then each bag has ☐ red and ☐ blue.

Factors of 30: _____

Factors of 45: _____

8 **Bus A leaves a stop every 8 minutes. Bus B leaves every 12 minutes. Both leave at 9:00 am.**

(a) They next leave together at ☐ : ☐ am.

(b) How many more times do they leave together after 9:00 am and before 11:00 am? ☐ times

Bus A: 9:00, 9:08, _____

Bus B: 9:00, 9:12, _____

9 **Who am I? I am between 50 and 100. Both 6 and 7 go into me exactly.**

I am ☐ .

Start here

Nearly there. You can do it.

10 **Count the factors of each number.**

Number	8	14	18	27	32	49
How many factors?						

(a) Write the number of factors under each number.

(b) Which number in the table has exactly 3 factors? ☐

11 **Mei buys pens in packs of 6 and rubbers in packs of 8. She wants the same number of pens and rubbers.**

(a) What is the fewest of each she can buy? ☐ of each

(b) How many packs of pens is that? ☐ packs

(c) How many packs of rubbers is that? ☐ packs

Multiples of 6: _____

Multiples of 8: _____

12 **I am a factor of 60. I am a multiple of 5. I am greater than 10 and less than 30. Write every number that fits.**

_____ _____

13 **Find out which numbers have an odd count of factors.**

(a) How many factors does 64 have? ☐

(b) How many factors does 81 have? ☐

(c) Which has an odd count of factors, 72 or 49? 72 49

Think harder

Hints are upside down. Try first!

1 **A number between 10 and 100 is in the 3 times table and in the 8 times table. Its tens digit is greater than its ones digit. Its digits add up to 9. What is the number?**

☐

Working: _____

_____

Hint. List the multiples of 24. Then use the digit clues.

2 **Ben has more than 50 stickers and fewer than 100. He packs them in 4s and 3 are left over. He packs them in 5s and 3 are left over. He packs them in 6s and 3 are left over. How many stickers does he have?**

☐

Working: _____

_____

_____

Hint. Take 3 away. What number is in the 4, 5 and 6 times tables?

3 **Two gears have marks that touch at the start. Gear A has 12 teeth. Gear B has 18 teeth. They turn together until the marks touch again.**

(a) How many teeth go past the touching point? ☐ teeth

(b) Gear A turns ☐ times and gear B turns ☐ times.

Working: _____

_____

_____

Hint. The number of teeth is a multiple of both 12 and 18.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **What is the least number that has exactly 10 factors? Show the factors you found.**

☐

Factors of 48: _____

Hint. Count factors of numbers that are 24, 30, 36, 40 and 48.

2 **Beyond Level 4: Three common multiples**

Three lighthouses flash at 12:00:00. Lighthouse A flashes every 6 seconds, B every 8 seconds and C every 9 seconds.

(a) They next flash together after ☐ seconds.

(b) How many times do they flash together in the next 10 minutes? Do not count 12:00:00. ☐ times

Hint. A common multiple of 3 numbers is in all 3 lists.

3 **Muffins are sold only in boxes of 4 and boxes of 7. Some numbers of muffins cannot be bought exactly. For example, you cannot buy exactly 5. What is the greatest number that cannot be bought exactly?**

☐

Boxes I can make: _____

Hint. Write the numbers 1 to 30 and cross out those you can make.

4 **A rectangle has an area of 84 cm². Its length and width are whole numbers of centimetres. Its perimeter is 38 cm. Find its length and its width.**

Length ☐ **cm** **Width** ☐ **cm**

Hint. Write the factor pairs of 84. Find the pair that adds to 19.

✔ Check yourself

No hints. Try each one on your own.

1 **Write all the factors of 52, in order.**

_____

Working: _____

2 **Siti swims every 6 days. Wei swims every 9 days. They both swim today. In how many days will they next swim on the same day?** ☐ **days**

Working: _____

3 **Kumar has 40 red pens and 64 blue pens. He makes identical sets with none left over. How many sets can he make at most?** ☐ **sets**

Working: _____

4 **I am a multiple of 7. I am a factor of 84. My digits add up to 5. Who am I?**

☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor Hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Factor Hunters

Your turn
page 6

Answer: 1, 3, 5, 9, 15, 45
Why: 1 × 45, 3 × 15 and 5 × 9 are the pairs. 2 and 4 do not go into 45. After 5 comes 9, which is already used, so stop.

START HERE

1
page 7

Answer: (a) 2 × 28, 4 × 14, 7 × 8 (b) 8 factors
Why: 3, 5 and 6 do not go into 56. The pairs 1 × 56, 2 × 28, 4 × 14 and 7 × 8 give 8 factors. The next number, 8, is already used.

2
page 7

Answer: 1, 2, 3, 6, 11, 22, 33, 66
Why: 1 × 66, 2 × 33, 3 × 22 and 6 × 11. Then 7, 8, 9 and 10 do not go into 66, and 11 is already used.

3
page 7

Answer: 2, 4, 8, 16
Why: Team sizes are factors of 32 other than 1 and 32: 2, 4, 8 and 16. For example, 4 children in a team gives 8 teams.

4
page 7

Answer: (a) True (b) False (c) True (d) True
Why: (a) 7 × 13 = 91. (b) The multiples of 8 are 48, 56, so 54 is not one. (c) 9 × 12 = 108. (d) 13 × 5 = 65.

5
page 7

Answer: (a) 9, 18, 27, 36, 45, 54 (b) 12, 24, 36, 48, 60, 72 (c) 36
Why: Count in 9s and in 12s. The first number in both lists is 36 (4 nines and 3 twelves).

6
page 8

Answer: (a) 1, 2, 4, 7, 14, 28 (b) 1, 2, 3, 6, 7, 14, 21, 42 (c) 1, 2, 7, 14 (d) 14
Why: 28 = 1 × 28, 2 × 14, 4 × 7. 42 = 1 × 42, 2 × 21, 3 × 14, 6 × 7. The numbers in both lists are 1, 2, 7 and 14. The greatest is 14.

7
page 8

Answer: (a) 15 bags (b) 2 red and 3 blue
Why: The number of bags is a common factor of 30 and 45. The common factors are 1, 3, 5 and 15, so the greatest is 15. Each bag has 30 ÷ 15 = 2 red and 45 ÷ 15 = 3 blue.

8
page 8

Answer: (a) 9:24 am (b) 4 times
Why: The least common multiple of 8 and 12 is 24, so they meet at 9:24 am. They meet every 24 minutes: 9:24, 9:48, 10:12 and 10:36. The next, 11:00, is not before 11:00. That is 4 times.

9
page 8

Answer: 84
Why: A number in both the 6 and 7 times tables: 42, 84. Only 84 is between 50 and 100.

10
page 9

Answer: (a) 8: 4, 14: 4, 18: 6, 27: 4, 32: 6, 49: 3 (b) 49
Why: 8: 1, 2, 4, 8. 14: 1, 2, 7, 14. 18: 1, 2, 3, 6, 9, 18. 27: 1, 3, 9, 27. 32: 1, 2, 4, 8, 16, 32. 49: 1, 7, 49. Only 49 has 3 factors.

11 page 9
Answer: (a) 24 (b) 4 (c) 3
Why: The least common multiple of 6 and 8 is 24 (6, 12, 18, 24 and 8, 16, 24). 24 ÷ 6 = 4 packs of pens and 24 ÷ 8 = 3 packs of rubbers.

12 page 9
Answer: 15, 20
Why: The factors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. Of these, 15 and 20 are multiples of 5 between 10 and 30.

13 page 9
Answer: (a) 7 (b) 5 (c) 49
Why: 64: 1, 2, 4, 8, 16, 32, 64 is 7 factors. 81: 1, 3, 9, 27, 81 is 5 factors. 49 = 7 × 7 pairs with itself, so it has 3 factors (odd). 72 has 12 factors.

THINK HARDER

1 page 10
Answer: 72
Why: A multiple of 3 and 8 is a multiple of 24: 24, 48, 72, 96. The digits add to 9 only for 72 (7 + 2). 7 is greater than 2.

2 page 10
Answer: 63
Why: Without the 3 left over, the number is in the 4, 5 and 6 times tables, so it is a multiple of 60. 60 + 3 = 63 is between 50 and 100. The next one, 123, is too big.

3 page 10
Answer: (a) 36 teeth (b) A 3 times and B 2 times
Why: The least common multiple of 12 and 18 is 36. Gear A: 36 ÷ 12 = 3 turns. Gear B: 36 ÷ 18 = 2 turns.

GO FURTHER

1 page 11
Answer: 48
Why: 48 has 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors. Numbers below 48 have fewer: for example 36 has 9, 40 has 8 and 42 has 8.

2 page 11
Answer: (a) 72 (b) 8 times
Why: The least common multiple of 6, 8 and 9 is 72. 10 minutes is 600 seconds. 72 × 8 = 576 is less than 600 and 72 × 9 = 648 is more, so they flash together 8 times.

3 page 11
Answer: 17
Why: List the amounts you can make: 4, 7, 8, 11, 12, 14, 15, 16, 18, 19, 20, 21, 22, 23 and every larger number. 17 is not on the list, and 17 − 7 = 10 and 17 − 14 = 3 are not multiples of 4. Every number from 18 up can be made.

4 page 11
Answer: Length 12 cm, width 7 cm
Why: Half the perimeter is 19, so length + width = 19. The factor pairs of 84 are 1 and 84, 2 and 42, 3 and 28, 4 and 21, 6 and 14, 7 and 12. Only 7 and 12 add to 19.

CHECK YOURSELF

1 page 12
Answer: 1, 2, 4, 13, 26, 52
Why: 1 × 52, 2 × 26 and 4 × 13. Then 5 to 12 do not go into 52, and 13 is already used.

2
page 12

Answer: 18
Why: Multiples of 6: 6, 12, 18. Multiples of 9: 9, 18. The first number in both lists is 18.

3
page 12

Answer: 8
Why: The factors of 40 and 64 that are in both lists are 1, 2, 4, 8. The greatest is 8 (5 red and 8 blue in each set).

4
page 12

Answer: 14
Why: The factors of 84 that are multiples of 7 are 7, 14, 21, 28, 42 and 84. Only 14 has digits that add up to 5 (1 + 4).